NVMe/FC Host Configuration for Windows Server 2012 R2 with ONTAP

ONTAP SAN Host

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NVMe/FC Host Configuration for Windows Server 2012 R2 with ONTAP

Supportability

NVMe/FC is supported on ONTAP 9.7 or later for Windows Server 2016.

Note that the Broadcom initiator can serve both NVMe/FC and FCP traffic through the same 32G FC adapter ports. For FCP and FC/NVMe, use MSDSM as the Microsoft Multipath I/O (MPIO) option.

See the Hardware Universe for a list of supported FC adapters and controllers. For the most current list of supported configurations & versions, see the NetApp Interoperability Matrix.

Known limitations

Windows Failover Cluster (WFC) is not supported with ONTAP NVMe/FC because ONTAP does not currently support persistent reservations with NVMe/FC.

> The external driver shipped by Broadcom for Windows NVMe/FC is not a true NVMe/FC driver but a translational SCSI □ NVMe driver. This translational overhead does not necessarily impact performance, but it does negate the performance benefits of NVMe/FC. Thus, on Windows servers, NVMe/FC and FCP performance is the same, unlike on other operating systems such as Linux, where NVMe/FC performance is significantly better than that of FCP.

Enabling NVMe/FC on a Windows initiator host

Follow these steps to enable FC/NVMe on the Windows initiator host:

1. Install OneCommand Manager utility on the Windows host.
2. On each of the HBA initiator ports, set the following HBA driver parameters:
   - EnableNVMe = 1
   - NVMEMode = 0
   - LimTransferSize=1
3. Reboot the host.

Configuring the Broadcom FC adapter in Windows for NVMe/FC

With the Broadcom adapter for FC/NVMe in a Windows environment, a `hostnqn` is associated with each host bus adapter (HBA) port. The `hostnqn` is formatted as follows.

Example

```
nqn.2017-01.com.broadcom:ecd:nvmf:fc:100000109b1b9765
nqn.2017-01.com.broadcom:ecd:nvmf:fc:100000109b1b9766
```

Enabling MPIO for NVMe devices on the Windows host

1. Install Windows Host Utility Kit 7.1 to set the driver parameters that are common to both FC and NVMe.
2. Open the MPIO properties.
3. From the **Discover Multi-Paths** tab, add the device ID listed for NVMe.

   MPIO becomes aware of the NVMe devices, which are visible under disk management.

4. Open **Disk Management** and go to **Disk Properties**.
5. From the **MPIO** tab, click **Details**.
6. Set the following MSDSM settings:
   - PathVerifiedPeriod: **10**
   - PathVerifyEnabled: **Enable**
   - RetryCount: **6**
   - RetryInterval: **1**
   - PDORemovedPeriod: **130**
7. Select the MPIO Policy **Round Robin with Subset**.
8. Change the registry values:

   ```
   HKLM\SYSTEM\CurrentControlSet\Services\mpio\Parameters\PathRecoveryInter
   val DWORD -> 30

   HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\
   UseCustomPathRecoveryInterval  DWORD-> 1
   ```

9. Reboot the host.

   The NVMe configuration is now complete on the Windows host.

Validating NVMe/FC

1. Validate that the Port Type is FC+NVMe.

   Now that NVMe is enabled, you should see the `Port Type` listed as `FC+NVMe`, as follows.

```
PS C:\Program Files\Emulex\Util\OCManager> .\hbacmd listhba

Manageable HBA List

Port WWN       : 10:00:00:10:9b:1b:97:65
Node WWN       : 20:00:00:10:9b:1b:97:65
Fabric Name    : 10:00:c4:f5:7c:a5:32:e0
Flags          : 8000e300
Host Name      : INTEROP-57-159
Mfg            : Emulex Corporation
Serial No.     : FC71367217
Port Number    : 0
Mode           : Initiator
PCI Bus Number : 94
PCI Function   : 0
Port Type      : FC+NVMe
Model          : LPe32002-M2

Port WWN       : 10:00:00:10:9b:1b:97:66
Node WWN       : 20:00:00:10:9b:1b:97:66
Fabric Name    : 10:00:c4:f5:7c:a5:32:e0
Flags          : 8000e300
Host Name      : INTEROP-57-159
Mfg            : Emulex Corporation
Serial No.     : FC71367217
Port Number    : 1
Mode           : Initiator
PCI Bus Number : 94
PCI Function   : 1
Port Type      : FC+NVMe
Model          : LPe32002-M2
```

2. Validate that NVMe/FC subsystems have been discovered.

   The `nvme-list` command lists the NVMe/FC discovered subsystems.

```
PS C:\Program Files\Emulex\Util\OCManager> .\hbacmd nvme-list
10:00:00:10:9b:1b:97:65

Discovered NVMe Subsystems for 10:00:00:10:9b:1b:97:65

NVMe Qualified Name     :  nqn.1992-
08.com.netapp:sn.a3b74c32db2911eab229d039ea141105:subsystem.win_nvme_int
erop-57-159
Port WWN                :  20:09:d0:39:ea:14:11:04
Node WWN                :  20:05:d0:39:ea:14:11:04
Controller ID           :  0x0180
Model Number            :  NetApp ONTAP Controller
Serial Number           :  81CGZBPU5T/uAAAAAAAB
Firmware Version        :  FFFFFFFF
Total Capacity          :  Not Available
Unallocated Capacity    :  Not Available

NVMe Qualified Name     :  nqn.1992-
08.com.netapp:sn.a3b74c32db2911eab229d039ea141105:subsystem.win_nvme_int
erop-57-159
Port WWN                :  20:06:d0:39:ea:14:11:04
Node WWN                :  20:05:d0:39:ea:14:11:04
Controller ID           :  0x0181
Model Number            :  NetApp ONTAP Controller
Serial Number           :  81CGZBPU5T/uAAAAAAAB
Firmware Version        :  FFFFFFFF
Total Capacity          :  Not Available
Unallocated Capacity    :  Not Available
Note: At present Namespace Management is not supported by NetApp Arrays.
```

```
PS C:\Program Files\Emulex\Util\OCManager> .\hbacmd nvme-list
10:00:00:10:9b:1b:97:66

Discovered NVMe Subsystems for 10:00:00:10:9b:1b:97:66

NVMe Qualified Name     :  nqn.1992-
08.com.netapp:sn.a3b74c32db2911eab229d039ea141105:subsystem.win_nvme_int
erop-57-159
Port WWN                :  20:07:d0:39:ea:14:11:04
Node WWN                :  20:05:d0:39:ea:14:11:04
Controller ID           :  0x0140
Model Number            :  NetApp ONTAP Controller
Serial Number           :  81CGZBPU5T/uAAAAAAAB
Firmware Version        :  FFFFFFFF
Total Capacity          :  Not Available
Unallocated Capacity    :  Not Available

NVMe Qualified Name     :  nqn.1992-
08.com.netapp:sn.a3b74c32db2911eab229d039ea141105:subsystem.win_nvme_int
erop-57-159
Port WWN                :  20:08:d0:39:ea:14:11:04
Node WWN                :  20:05:d0:39:ea:14:11:04
Controller ID           :  0x0141
Model Number            :  NetApp ONTAP Controller
Serial Number           :  81CGZBPU5T/uAAAAAAAB
Firmware Version        :  FFFFFFFF
Total Capacity          :  Not Available
Unallocated Capacity    :  Not Available

Note: At present Namespace Management is not supported by NetApp Arrays.
```

3. Validate that namespaces have been created.

   The `nvme-list-ns` command lists the namespaces for a specified NVMe target that lists the namespaces connected to the host.

```
PS C:\Program Files\Emulex\Util\OCManager> .\HbaCmd.exe nvme-list-ns
10:00:00:10:9b:1b:97:66 20:08:d0:39:ea:14:11:04 nq
.1992-
08.com.netapp:sn.a3b74c32db2911eab229d039ea141105:subsystem.win_nvme_int
erop-57-159 0


Active Namespaces (attached to controller 0x0141):

                                        SCSI           SCSI
SCSI
   NSID           DeviceName        Bus Number    Target Number     OS
LUN
-----------  --------------------  ------------  ---------------
---------
0x00000001   \\.\PHYSICALDRIVE9         0               1              0
0x00000002   \\.\PHYSICALDRIVE10        0               1              1
0x00000003   \\.\PHYSICALDRIVE11        0               1              2
0x00000004   \\.\PHYSICALDRIVE12        0               1              3
0x00000005   \\.\PHYSICALDRIVE13        0               1              4
0x00000006   \\.\PHYSICALDRIVE14        0               1              5
0x00000007   \\.\PHYSICALDRIVE15        0               1              6
0x00000008   \\.\PHYSICALDRIVE16        0               1              7
```

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